Boolean Algebra Solutions – Intermediate Division

1. **00-01 C3 Boolean Algebra**

$A(A+BC)+B(B+AC)=AA+ABC+BB+ABC$
$=A+ABC+B=A(1+BC)+B=A(1)+B=A+B$ $\quad A+B$

2. **01-02 C2 Boolean Algebra**

$(\overline{X}+Y)(\overline{X}+\overline{Y})=\overline{X}\,\overline{X}+\overline{X}\overline{Y}+Y\overline{X}+Y\overline{Y}=$
$\overline{X}+\overline{X}\overline{Y}+\overline{X}Y+0=\overline{X}(1+\overline{Y}+Y)=\overline{X}(1)=\overline{X}$ $\quad \overline{X}$

3. **01-02 C2 Boolean Algebra**

$\overline{AB}+A(\overline{B+C})=\overline{A}+\overline{B}+A(\overline{B}\,\overline{C})=\overline{A}+\overline{B}+A\overline{B}\,\overline{C}=\overline{A}+\overline{B}$ $\quad$ (1,1,0) and (1,1,1)

Therefore, $\overline{\overline{A}+\overline{B}}=AB$ makes it FALSE. That is (1,1,*).

4. **01-02 C3 Boolean Algebra**

$\overline{A}\;B+\overline{A}\;\overline{C}+\overline{A}\;\overline{B}+\overline{B}\;C+A\;\overline{C}+\overline{B}\;\overline{C}=$ $\quad \overline{ABC}$
$(\overline{A}\;B+\overline{A}\;\overline{B})+(\overline{B}\;C+\overline{B}\;\overline{C})+(A\;\overline{C}+\overline{A}\;\overline{C})=$
$\overline{A}\,(B+\overline{B})+\overline{B}\,(C+\overline{C})+\overline{C}\,(A+\overline{A})=$
$\overline{A}+\overline{B}+\overline{C}=\overline{ABC}$ $\quad$ (*note that* $\overline{A}\;\overline{B}\;\overline{C}$ *is not equivalent)*

5. **02-03 C2 Boolean Algebra**

B AND A OR A AND (NOT B OR NOT C) = $\quad \sim AB$
$AB+A(\overline{B}+\overline{C})=AB+A\overline{B}+A\overline{C}=$
$A(B+\overline{B})+A\overline{C}=A(1)+A\overline{C}=A\overline{C}$

6. **02-03 C2 Boolean Algebra**

NOT A AND C OR B AND (NOT (A OR C)) $\quad$ (0,1,0), (0,0,1), (0,1,1)

A	B	C	$\overline{A}$	$\overline{A}C$	$A+C$	$\overline{A+C}$	$B\overline{A+C}$	OR
0	0	0	1	0	0	1	0	0
0	0	1	1	1	1	0	0	1
0	1	0	1	0	0	1	1	1
0	1	1	1	1	1	0	0	1
1	0	0	0	0	1	0	0	0
1	0	1	0	0	1	0	0	0
1	1	0	0	0	1	0	0	0
1	1	1	0	0	1	0	0	0

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7. **02-03 C3 Boolean Algebra**

$\overline{\overline{(A+B)}(C\overline{A})}+\overline{\overline{B}A(\overline{B}+C)}=A+B+(\overline{C}\overline{\overline{A}})+\overline{\overline{B}}A+\overline{\overline{B}+C}=$
$A+B+\overline{C}+A+B+\overline{A}+B+\overline{C}=A+\overline{A}+B+\overline{C}=1+B+\overline{C}=1$

1

8. **03-04 C2 Boolean Algebra**

(0,1), (1,0)

A	B	AB	$\overline{A}\overline{B}$	+	NOT
0	0	0	1	1	0
0	1	0	0	0	1
1	0	0	0	0	1
1	1	1	0	1	0

9. **03-04 C2 Boolean Algebra**

$B(\overline{A}+B)+A(\overline{B}+C)+\overline{C}(A+B)=\overline{A}B+BB+A\overline{B}+AC+A\overline{C}+B\overline{C}$
$=\overline{A}B+B+A\overline{B}+A(1)+B\overline{C}=\overline{A}B+B+A\overline{B}+A+B\overline{C}=$
$=B(\overline{A}+1+\overline{C})+A(\overline{B}+1)=B+A$

$A+B$

10. **03-04 C3 Boolean Algebra**

$(A+\overline{B})(\overline{AB})\ (\overline{\overline{A}+B})=(A+\overline{B})(\overline{A}+\overline{B})(A\overline{B})$
$=(A+\overline{B})(A\overline{A}\overline{B}+A\overline{B})=(A+\overline{B})(A\overline{B})=A\overline{B}+A\overline{B}=A\overline{B}$

$A\overline{B}$

11. **04-05 C2 Boolean Algebra**

$\overrightarrow{(A+B)}\ (A\overline{B}+B\overline{C})=\overline{A}\overline{B}\ (A\overline{B}+B\overline{C})=$
$A\overline{A}\overline{B}+\overline{A}B\overline{B}\overline{C}=0+0=0$

0

12. **04-05 C2 Boolean Algebra**

$\overline{A}(B+\overline{C})+A\overline{B}C$

4

A	B	C	$\overline{A}$	$\overline{B}$	$\overline{C}$	$\overline{A}(B+\overline{C})$	$A\overline{B}C$	$\overline{A}(B+\overline{C})+A\overline{B}C$
0	0	0	1	1	1	1	0	1
0	0	1	1	1	0	0	0	0
0	1	0	1	0	1	1	0	1
0	1	1	1	0	0	1	0	1
1	0	0	0	1	1	0	0	0
1	0	1	0	1	0	0	1	1
1	1	0	0	0	1	0	0	0
1	1	1	0	0	0	0	0	0

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13. **04-05 C3 Boolean Algebra**

$\overline{A+\overline{B}}\quad \overline{C\overline{A}(B+C)} = (\overline{A}B)(\overline{C\overline{A}}+\overline{B+C}) =$ $\overline{A}B\overline{C}$

$\overline{A}B(\overline{C}+A+\overline{B}\overline{C}) = \overline{A}B\overline{C}+\overline{A}BA+\overline{A}B\overline{B}\overline{C} = \overline{A}B\overline{C}$

14. **05-06 C2 Boolean Algebra**

$(\overline{A+\overline{B}})(A\overline{B}+C)(B+\overline{C}) = \overline{A}B(A\overline{B}+C)(B+\overline{C}) =$ $\bar{A}BC$

$\overline{A}B(A\overline{B}B+A\overline{B}\overline{C}+CB+\overline{C}C) = \overline{A}B(A\overline{B}\overline{C}+CB) =$

$\bar{A}BA\bar{B}\bar{C}+\bar{A}BC = \bar{A}BC$

15. **05-06 C2 Boolean Algebra**

$(\overline{A}+\overline{B})\overline{C}+\overline{A}(B\overline{C}+A)+(\overline{C+A})(\overline{A}B) = \overline{A}\,\overline{C}+\overline{B}\overline{C}+\overline{A}B\overline{C}+\overline{A}A+\overline{A}B\overline{C} =$ 3

$\overline{A}\,\overline{C}+\overline{B}\overline{C}+\overline{A}B\overline{C} = \overline{A}\,\overline{C}(1+B)+\overline{B}\overline{C} = \overline{A}\,\overline{C}+\overline{B}\overline{C} = \overline{C}(\overline{A}+\overline{B}) = 1$

Both factors must =1. Therefore $\overline{C} = 1 \Rightarrow C = 0$

$(\overline{A}+\overline{B}) = 1 \Rightarrow (0,0)$ OR $(0,1)$ OR $(1,0)$

The ordered triples are: (0,0,0), (0,1,0) and (1,0,0)

16. **05-06 C3 Boolean Algebra**

$\overline{\overline{\overline{A}+AB}(B+C)C} = \overline{\overline{\overline{A}}\,\overline{AB}(BC+C)} = \overline{A(\overline{A}+\overline{B})(BC+C)} =$ $\overline{A\overline{B}C}$

$\overline{(A\overline{A}+A\overline{B})C(B+1)} = \overline{A\overline{B}C}$

17. **06-07 C2 Boolean Algebra**

$\overline{A}(B+\overline{A})+\overline{A}(\overline{B}+A) = \overline{A}B+\overline{A}\,\overline{A}+\overline{A}\,\overline{B}+\overline{A}A = \overline{A}B+\overline{A}+\overline{A}\,\overline{B} =$ $\overline{A}$

$\overline{A}(B+1+\overline{B}) = \overline{A}$

18. **06-07 C2 Boolean Algebra**

The expression simplifies to: $A\overline{B}+AC+\overline{A}\,\overline{B}C$ (0,0,1),(1,0,0),(1,0,1),(1,1,1)

A	B	C	$A\overline{B}$	AC	$\overline{A}\,\overline{B}C$	+
0	0	0	0	0	0	0
0	0	1	0	0	1	1
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	1	0	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	0
1	1	1	0	1	0	1

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19. **06-07 C3 Boolean Algebra**

$AB\overline{C} + A\overline{B}\,\overline{C} + \overline{A}BC + \overline{A}\,\overline{B}C = A\overline{C}(B+\overline{B}) + \overline{A}C(B+\overline{B}) =$ (1,1,0), (1,0,0)

$A\overline{C} + \overline{A}C = A \oplus C \Rightarrow$ A=1, B = *, C = 0 OR A = 0, B = * C = 1 (0,1,1), (0,0,1)

20. **07-08 C2 Boolean Algebra**

$\overline{A}(AB+C) + B(\overline{A}C + \overline{B})$ $= \overline{A}AB + \overline{A}C + B\overline{A}C + B\overline{B}$ $\overline{A}C$

$= 0 + \overline{A}C + B\overline{A}C + 0$

$= \overline{A}C(1+B)$

$= \overline{A}C$

21. **07-08 C2 Boolean Algebra**

$\overline{A}(A+B\overline{C}) + B\overline{C}(A+\overline{B})$ $= \overline{A}A + \overline{A}B\overline{C} + AB\overline{C} + B\overline{B}C$ (1,1,0), (0,1,0)

$= \overline{A}B\overline{C} + AB\overline{C}$

$= B\overline{C}(\overline{A}+A)$

$= B\overline{C}$

If $B\overline{C} = 1$, then $B = 1$ and $C = 0$. Also $A =$ *.

22. **07-08 C3 Boolean Algebra**

$(A \oplus B)(A+BC)$ 3

A	B	C	$A \oplus B$	BC	$A+BC$	$(A \oplus B)(A+BC)$
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	1	0	0	0
0	1	1	1	1	1	1
1	0	0	1	0	1	1
1	0	1	1	0	1	1
1	1	0	0	0	1	0
1	1	1	0	1	1	0

23. **08-09 C2 Boolean Algebra**

$AB(\overline{\overline{B}+C}) + \overline{C}(\overline{AC+\overline{B}}) = AB(\overline{\overline{B}+C})\overline{C}(\overline{AC+\overline{B}})$ ABC

$= AB(\overline{B}+C)\overline{\overline{C}(\overline{AC+\overline{B}})} = AB(\overline{B}+C)(C+AC+\overline{B})$

$= (AB\overline{B} + ABC)(AC+\overline{B}+C) = ABC + 1 = ABC$

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24. **08-09 C2 Boolean Algebra**

$A(\overline{A\overline{B}C + \overline{A}B\overline{C}})(\overline{A(\overline{B}C + A\overline{C})}) = 1 \quad \Rightarrow$ (1,1,1)

$A = 1 \wedge (\overline{A\overline{B}C + \overline{A}B\overline{C}}) = 1 \wedge (\overline{A(\overline{B}C + A\overline{C})}) = 1$

$(\overline{A\overline{B}C + \overline{A}B\overline{C}}) = 1 \Rightarrow A\overline{B}C + \overline{A}B\overline{C} = 0 \Rightarrow 1\overline{B}C + 1B\overline{C} = 0 \Rightarrow \overline{B}C + 0 = 0 \Rightarrow \overline{B}C = 0$

$(\overline{A(\overline{B}C + A\overline{C})}) = 1 \Rightarrow A(\overline{B}C + A\overline{C}) = 0 \Rightarrow \overline{B}C + \overline{C} = 0 \Rightarrow \overline{B}C = 0 \wedge \overline{C} = 0 \Rightarrow C = 1$

$\overline{B}C = 0 \wedge C = 1 \Rightarrow \overline{B} = 0 \Rightarrow B = 1.$ Therefore (1,1,1) is the only solution.

Note: using a table might provide an easier solution.

25. **08-09 C3 Boolean Algebra**

$(A + BC)(\overline{\overline{A}B + \overline{C}}) = (A + BC)(\overline{\overline{A}}\,\overline{B}\overline{C}) = (A + BC)((\overline{\overline{A}} + \overline{B})C)$ AC

$= (A + BC)(AC + \overline{B}C) = AC + A\overline{B}C + ABC + B\overline{B}C$

$= AC(1 + \overline{B} + B) = AC$

26. **09-10 C2 Boolean Algebra**

$(A\overline{B} + B\overline{C})(\overline{A}C + B) = A\overline{A}\overline{B}C + AB\overline{B} + \overline{A}BC\overline{C} + BB\overline{C}$ $B\overline{C}$

$= 0 + 0 + 0 + B\overline{C}$

$= B\overline{C}$

27. **09-10 C2 Boolean Algebra**

$\overline{\overline{A(\overline{B} + C)} + \overline{A}B + C} = 1 \quad \Rightarrow \overline{A(\overline{B} + C)} + \overline{A}B + C = 0$ (1,0,0)

$\Rightarrow \overline{A} + \overline{\overline{B} + C} + \overline{A}B + C$

$\Rightarrow \overline{A} + B\overline{C} + \overline{A}B + C = 0$

So $C = 0 \wedge \overline{A} = 0 \Rightarrow A = 1.$ Then $0 + B*1 + 0*B + 0 = 0 \Rightarrow B = 0$

28. **09-10 C3 Boolean Algebra**

$\overline{(AB + \overline{C}) + (A(\overline{B} + C))}$

$= \overline{(AB + \overline{C})}\,\overline{A(\overline{B} + C)}$

$= \overline{AB}\,C(\overline{A} + \overline{\overline{B} + C})$ $\overline{A}C$

$= (\overline{A} + \overline{B})C(\overline{A} + B\overline{C})$

$= \overline{A}C + \overline{A}B\overline{C}C + \overline{A}\overline{B}C + B\overline{B}C\overline{C}$

$= \overline{A}C$

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29. **10-11 C2 Boolean Algebra**

$A\overline{B}(B+C)+\overline{A(\overline{B+C})(B+\overline{C})} = A\overline{B}B + A\overline{B}C + \overline{A} + \overline{\overline{B+C}} + \overline{B+\overline{C}}$ (1,0,0)

$= 0 + A\overline{B}C + \overline{A} + B + C + \overline{B}C = C(A\overline{B}+1+\overline{B}) + \overline{A} + B = \overline{A} + B + C$

If $\overline{A}+B+C = 0$, then $\overline{A} = 0 \wedge B = 0 \wedge C = 0$. So $A = 1 \wedge B = 0 \wedge C = 0$

30. **10-11 C2 Boolean Algebra**

$A(\overline{A\overline{B}+BC}) = A((\overline{A\overline{B}})(\overline{BC})) = A((\overline{A}+B)(\overline{B}+\overline{C})) = (A\overline{A}+AB)(\overline{B}+\overline{C})$ $AB\overline{C}$

$= (0+AB)(\overline{B}+\overline{C}) = AB\overline{B} + AB\overline{C} = 0 + AB\overline{C} = AB\overline{C}$

31. **10-11 C3 Boolean Algebra**

$A(\overline{A}B+C)+\overline{\overline{B}(A+BC)}+C(\overline{A}B+\overline{B}C)$ 1

$= A\overline{A}B + AC + \overline{A\overline{B} + B\overline{B}C} + \overline{A}BC + \overline{B}C$

$= 0 + AC + \overline{A\overline{B}} + \overline{A}BC + \overline{B}C$

$= AC + \overline{A} + B + \overline{A}BC + \overline{B}C$

If $AC + \overline{A} + B + \overline{A}BC + \overline{B}C = 0$, then $AC = 0 \wedge \overline{A} = 0 \wedge B = 0 \wedge \overline{A}BC = 0 \wedge \overline{B}C = 0$

So $C = 0 \wedge A = 1 \wedge B = 0$. Therefore only 1 triple makes it FALSE.

32. **11-12 C3 Boolean Algebra**

$\overline{AB} + B(\overline{A+C}) = \overline{A} + \overline{B} + B(\overline{A}\,\overline{C})$ $\overline{A}+\overline{B}$ or $\overline{AB}$

$= \overline{A} + \overline{B} + \overline{A}B\overline{C}$

$= \overline{A}(1+B\overline{C}) + \overline{B} = \overline{A} + \overline{B}$ or $\overline{AB}$

33. **11-12 C3 Boolean Algebra**

$\overline{A}(B+C) + (\overline{A+C})(\overline{BC}) = \overline{A}B + \overline{A}C + \overline{A}\,\overline{C}(\overline{B}+\overline{C})$ 4

$= \overline{A}B + \overline{A}C + \overline{A}\,B\,\overline{C} + \overline{A}\,\overline{C}$

$= \overline{A}B + \overline{A}\,B\,\overline{C} + \overline{A}(C+\overline{C})$

$= \overline{A}B + \overline{A}\,B\,\overline{C} + \overline{A}$

$= \overline{A}(B + B\,\overline{C} + 1) = \overline{A}$ So 4 make it TRUE : (0,*,*)

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34. **12-13 C3 Boolean Algebra**

$(\overline{A}B+\overline{B})(A+B)(A\overline{B}+\overline{A})=(\overline{A}BA+\overline{A}BB+A\overline{B}+\overline{B}B)(A\overline{B}+\overline{A})$ $A\oplus B$

$=(\overline{A}B+A\overline{B})(A\overline{B}+\overline{A})=\overline{A}BA\overline{B}+\overline{A}\,\overline{A}B+A\overline{B}+A\overline{A}\,\overline{B}=A\oplus B$

35. **12-13 C3 Boolean Algebra**

$(A+\overline{B}C)(BC+\overline{A})+A(B+\overline{C})(\overline{A}+\overline{B}\,\overline{C})$ 3

$= ABC+A\overline{A}+\overline{B}CBC+\overline{A}\overline{B}C+AB\overline{A}+AB\overline{B}\,\overline{C}+A\overline{C}\overline{A}+A\overline{C}\overline{B}\overline{C}$

$= ABC+\overline{A}\overline{B}C+A\overline{B}\overline{C}$. That gives you (1,1,1), (0,0,1), and (1,0,0).

36. **13-14 C3 Boolean Algebra**

$(A\overline{B}+B\overline{C})(\overline{A}B\overline{C}+A\overline{B}C)=A\overline{B}\overline{A}B\overline{C}+A\overline{B}A\overline{B}C+B\overline{C}\overline{A}B\overline{C}+B\overline{C}A\overline{B}C$ (1,0,1), (0,1,0)

$=A\overline{A}B\overline{B}\overline{C}+AA\overline{B}\overline{B}C+ABB\overline{C}\overline{C}+AB\overline{B}C\overline{C}$

$=0+A\overline{B}C+\overline{A}B\overline{C}+0=A\overline{B}C+\overline{A}B\overline{C}$

$A=1\Rightarrow\overline{B}C=1\Rightarrow B=0\wedge C=1$ and $A=0\Rightarrow B\overline{C}=1\Rightarrow B=1\wedge C=0$

Therefore (1, 0, 1) and (0, 1, 0) make it true.

37. **13-14 C3 Boolean Algebra**

$A\overline{B}+B(C+\overline{A})+B\overline{C}+AB\overline{C}=A\overline{B}+BC+\overline{A}B+B\overline{C}+AB\overline{C}$ 6

$= A\overline{B}+BC+\overline{A}B+B\overline{C}(1+A)= A\overline{B}+BC+\overline{A}B+B\overline{C}$

$= A\overline{B}+B(C+\overline{C})+\overline{A}B = A\overline{B}+B+\overline{A}B=A\overline{B}+B(1+\overline{A})=A\overline{B}+B$

So $C=*$ $B=1\Rightarrow A*0+1=1\Rightarrow A=*$ $\therefore(*,1,*)$

$B=0\Rightarrow A*1+0=A\Rightarrow A=1$ $\therefore(1,0,*)$

That is (1,1,1), (1,1,0), (0,1,1), (0,1,0), (1,0,1),(1,0,0) with no duplicates.

38. **14-15 C3 Boolean Algebra**

$(A\overline{B}+AC+B\overline{C})(B\overline{C}+\overline{A}B+\overline{A}C)$ $B\overline{C}$

$=A\overline{B}B\overline{C}+A\overline{B}\overline{A}B+A\overline{B}\overline{A}C+ACB\overline{C}+AC\overline{A}B+AC\overline{A}C+B\overline{C}B\overline{C}+B\overline{C}\overline{A}B+B\overline{C}\overline{A}C$

$=0+0+0+0+0+0+B\overline{C}+\overline{A}B\overline{C}+0$

$= B\overline{C}(1+\overline{A})=B\overline{C}$

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39. **14-15 C3 Boolean Algebra**

$\overline{A}\overline{B}(\overline{\overline{A}+B+C})(\overline{A+B\overline{C}}) = (\overline{A}+\overline{B})(A\overline{B}\overline{C})(\overline{A}\overline{\overline{B}\overline{C}})$ 8

$= (\overline{A}A\overline{B}\overline{C}+\overline{B}A\overline{B}\overline{C})(\overline{A}(\overline{B}+C))$

$= A\overline{B}\overline{C}(\overline{A}\overline{B}+\overline{A}C) = A\overline{B}\overline{C}\overline{A}\overline{B}+A\overline{B}\overline{C}\overline{A}C = 0+0 = 0$

Therefore this circuit is always false.

40. **15-16 C3 Boolean Algebra**

$\overline{A}(B+A\overline{C})+B(C+\overline{A}B) = \overline{A}B+\overline{A}A\overline{C}+BC+B\overline{A}B$ (1,1,1), (0,1,1), (0,1,0)

$= \overline{A}B+0+BC+\overline{A}B = \overline{A}B+BC = B(\overline{A}+C)$

If $B(\overline{A}+C)=1$, then $B=1\wedge\overline{A}+C=1$

If $C=1$, then $A=*$. If $C=0$, then $\overline{A}=1$. So $A=0$.

41. **15-16 C3 Boolean Algebra**

$\overline{A}(\overline{B+A\overline{C}})+\overline{B\overline{C}}(\overline{\overline{A}+C}) = \overline{A}(\overline{B}\overline{A\overline{C}})+(\overline{B}+\overline{C})(\overline{\overline{A}\overline{C}})$ 4

$= \overline{A}\overline{B}(\overline{A}+\overline{\overline{C}})+A\overline{B}\overline{C}+A\overline{C}\overline{C} = \overline{A}\overline{A}\overline{B}+\overline{A}\overline{B}C+A\overline{B}\overline{C}+A\overline{C}$

$= \overline{A}\overline{B}(1+C)+A\overline{C}(\overline{B}+1) = \overline{A}\overline{B}+A\overline{C}$

If $A=0$, then $B=0\wedge C=*$. If $A=1$, then $C=0\wedge B=*$. Thus 4 triples.